ME 325 – Mechanical Engineering Group Project

Conceptual Design and Bench-scale Demonstration of a Hybrid Powertrain for a a Small Urban Vehicle

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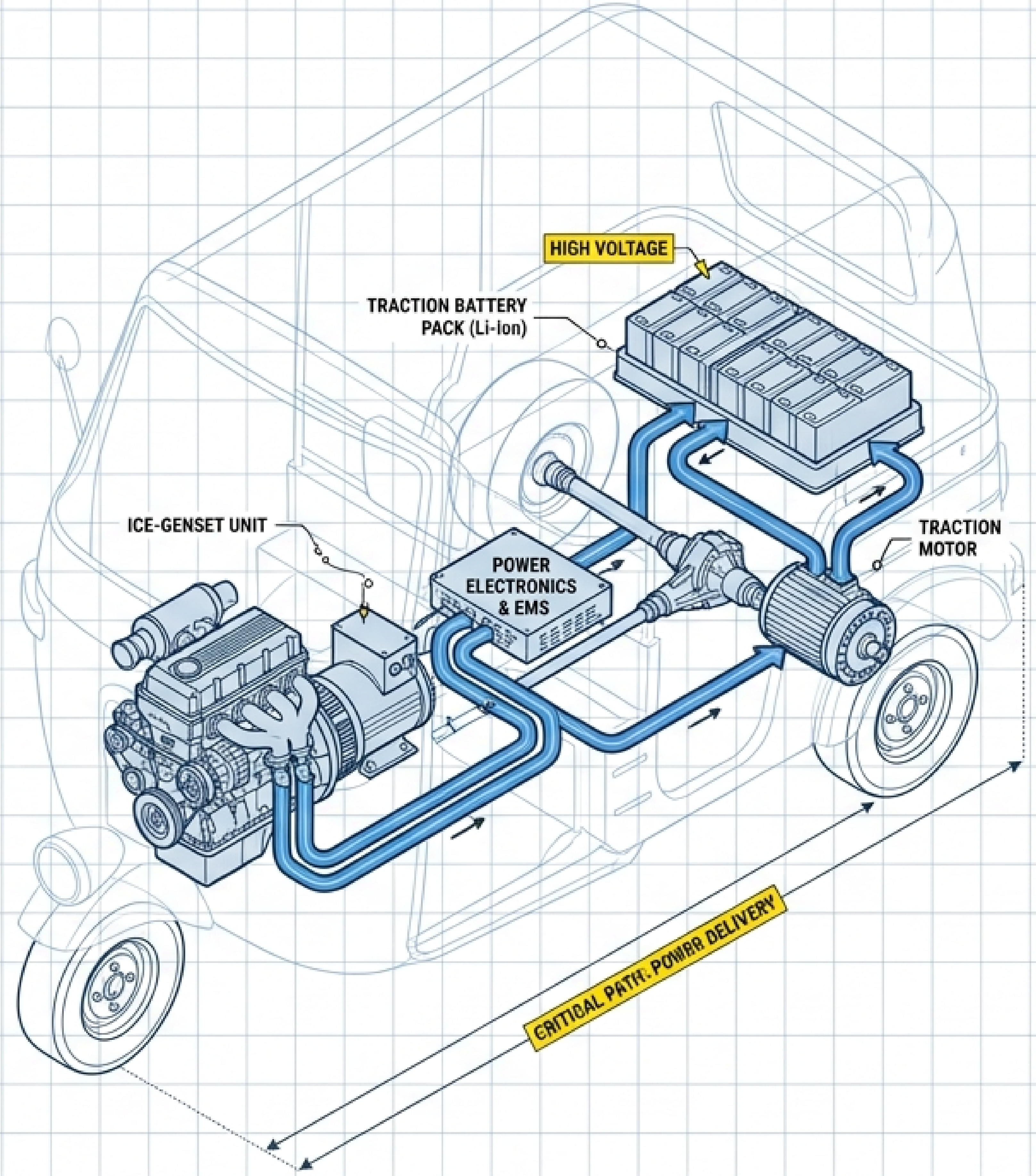


Small urban vehicles spend 87.25% of operational life in transient or idle states, making steady-state ICE architectures highly inefficient.

The Colombo Drive Cycle Reality

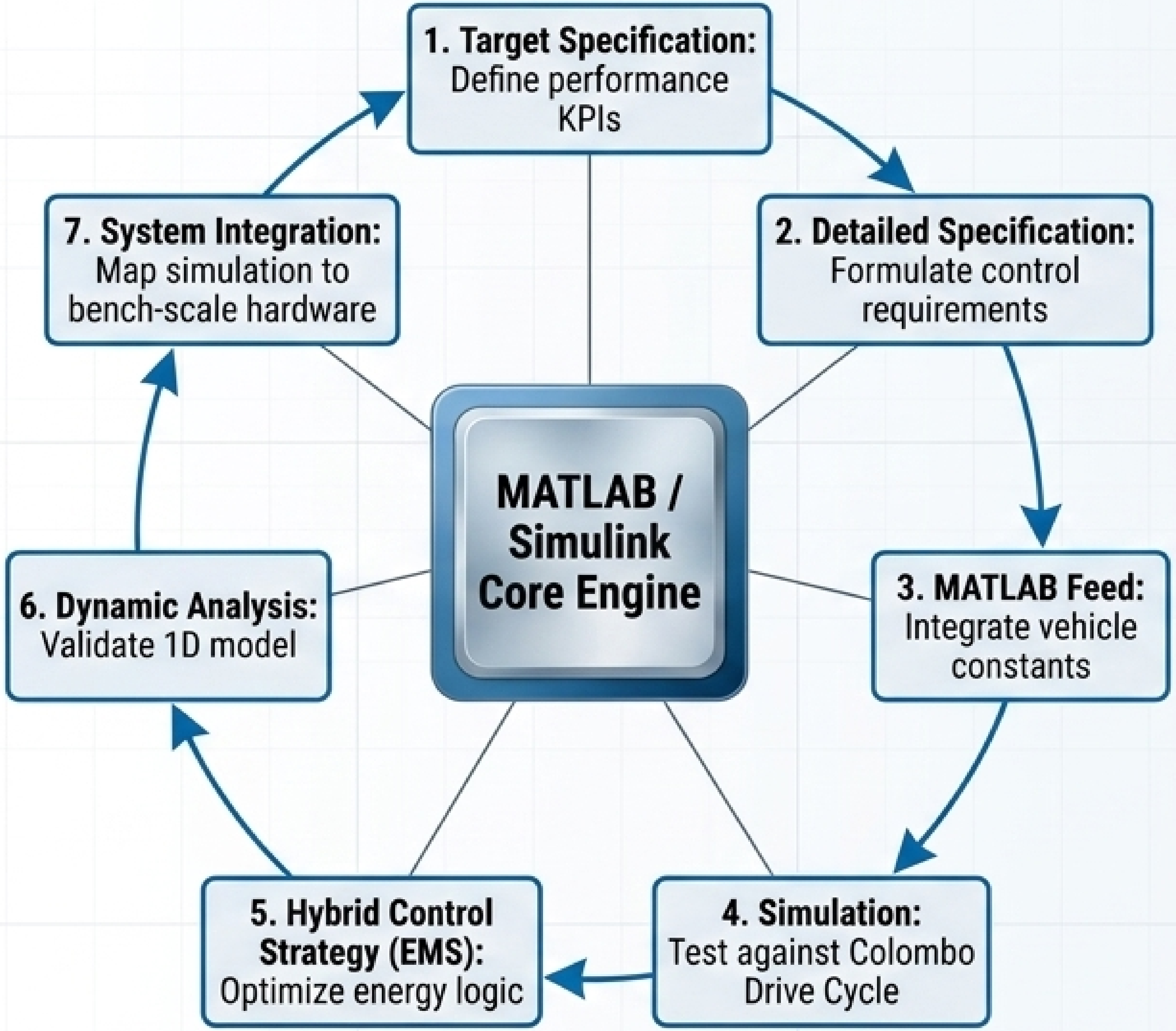
Average Speed: 20.3 km/h

- **20.5%** Idle Time — Extreme fuel waste while stationary.
- **36.1%** Acceleration — Demands high peak torque.
- **30.65%** Deceleration — Massive kinetic energy recovery potential.
- **12.75%** Cruise Mode — Minimal opportunity for direct engine-to-wheel drive.

Conventional ICE (200cc-450cc)	Pure Electric Vehicle (EV)	Proposed Series Hybrid
Low initial cost.	Zero tailpipe emissions.	Decouples engine from road speed for optimal operation.
High fuel waste during 20.5% idle.	HIGH initial cost.	Maximizes regenerative braking (no complex clutch).
0% braking recovery.	Long charging downtime causes lost daily revenue.	Converts idle time to battery charging.

Aim: Design a Series Hybrid Powertrain bridge to optimize fuel consumption without requiring immediate full electrification infrastructure.

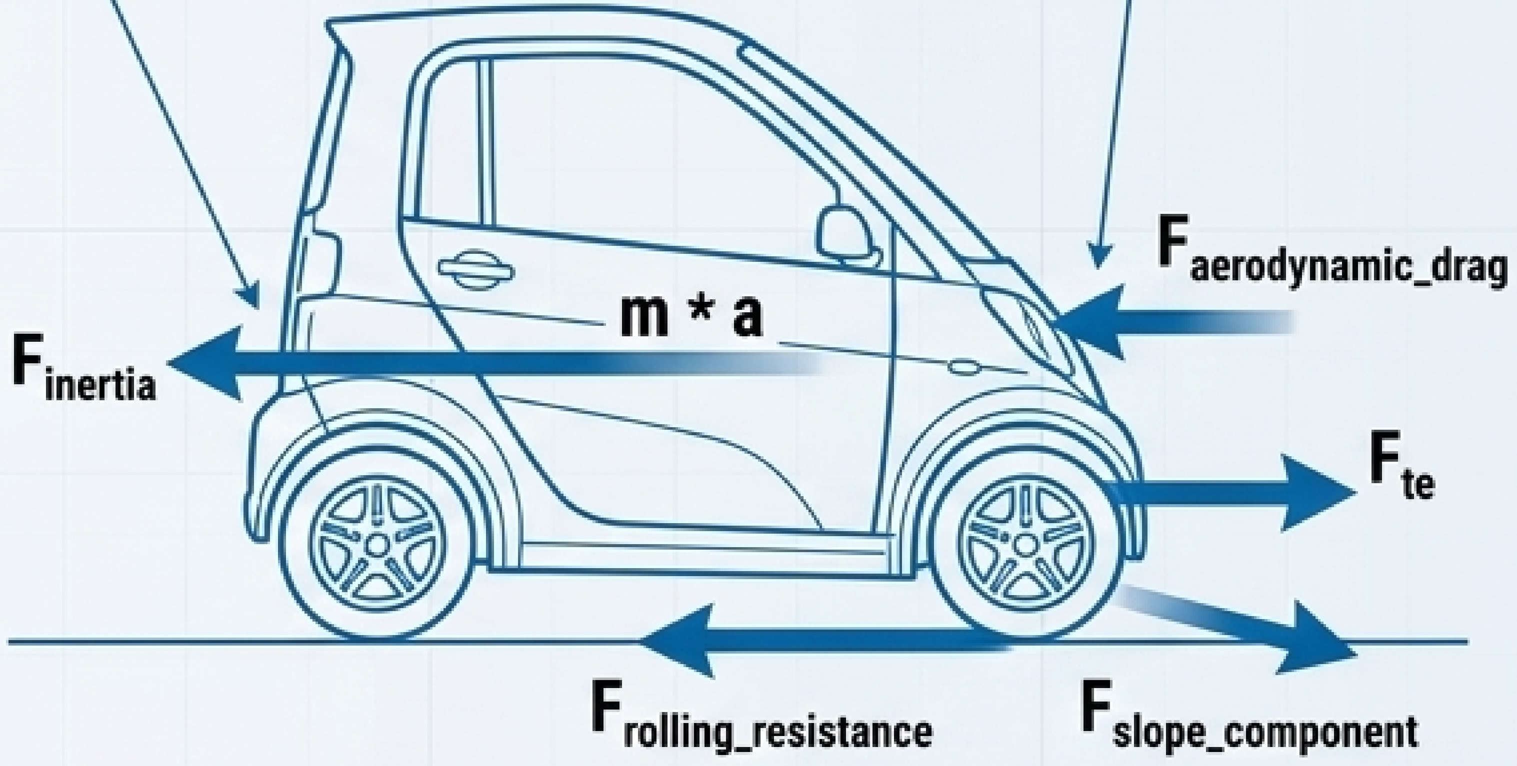
1D System Modeling and Longitudinal Dynamic Simulation



$$F_{te} = F_{inertia} + F_{rolling\_resistance} + F_{aerodynamic\_drag} + F_{slope\_component}$$

The primary hurdle during the 36.1% acceleration phase.

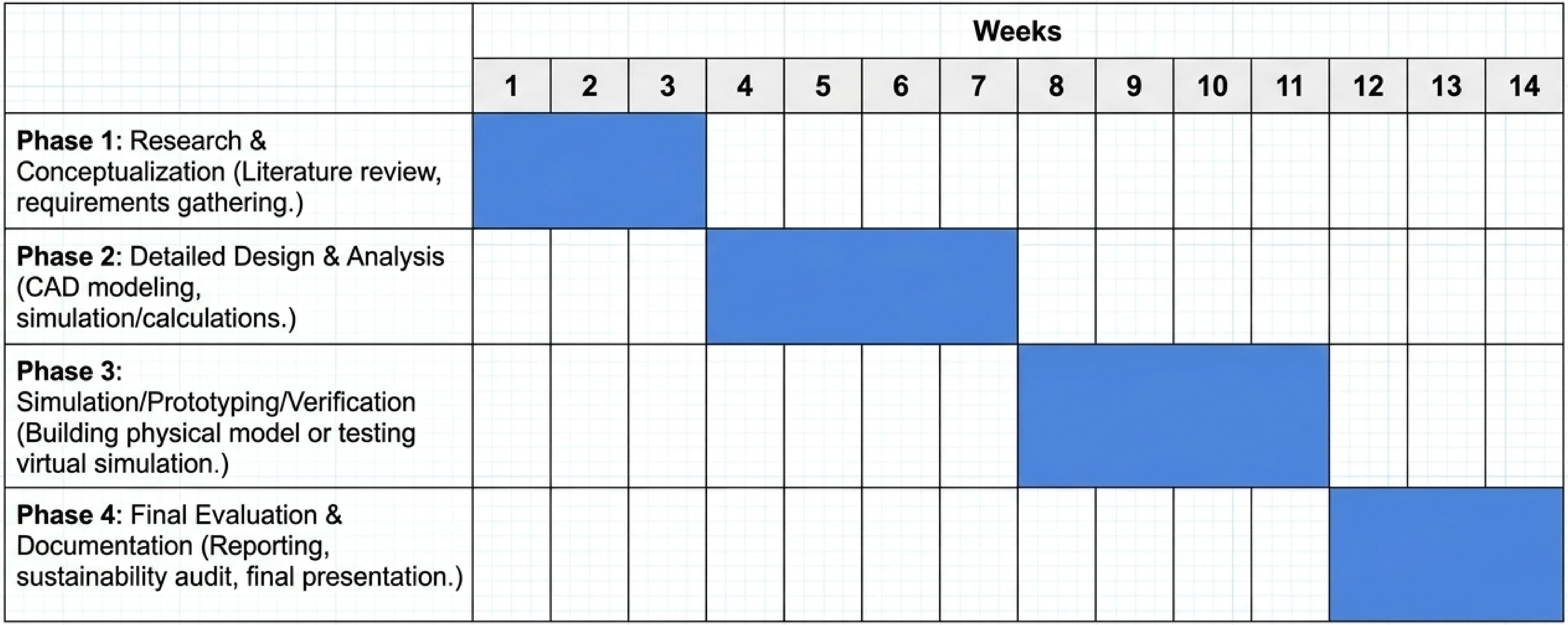
Negligible at the average speed of 20.3 km/h.



Key Insight: Series traction motor provides peak torque at 0 RPM to overcome high urban $F_{inertia}$.

14-Week Development Timeline and Execution Milestones

CRITICAL PATH: Successful integration of Rule-Based Energy Management Strategy (EMS) code with physical bench-scale hardware.



Systemic Impact of the Series Hybrid Architecture

Economic

Metric: 30% reduction in fuel expenditure.

Mechanism: Mitigates fuel waste during the 20.5% idle time in urban traffic.

End-User Benefit: Increases daily revenue margins for commercial operators by avoiding ICE idle costs and EV charging downtime.

Social

Metric: Noticeable reduction in localized urban pollution.

Mechanism: Automatic engine shut-off at idle and optimal-RPM operation.

End-User Benefit: Improves urban public health by cutting noise and harmful emissions at street level during stop-and-go traffic.

Environmental

Metric: Recaptures 30.65% of drive cycle energy.

Mechanism: High-efficiency regenerative braking and engine downsizing.

End-User Benefit: Enhances sustainability by reducing raw material footprints and converting lost kinetic energy into battery storage.